动态随机一般均衡模型：理论与应用
第 7 章：两国开放经济

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A 动态系统

- 家庭部门行为方程

$$\beta\mathbb{E}_t\frac{C_{t+1}^{-\sigma}}{C_t^{-\sigma}}\frac{1}{\Pi_{t+1}}=\frac{1}{R_t},\quad \beta\mathbb{E}_t\frac{(C_{t+1}^*)^{-\sigma}}{(C_t^*)^{-\sigma}}\frac{1}{\Pi_{t+1}^*}=\frac{1}{R_t^*}$$

$$C_t^\sigma N_t^\varphi=\hat{W}_t,\quad (C_t^*)^\sigma(N_t^*)^\varphi=\hat{W}_t^*$$

$$(1-\alpha)\hat{P}_{Ht}^{1-\eta}+\alpha\hat{P}_{Ft}^{1-\eta}=1$$

$$(1-\alpha)(\hat{P}_{Ft}^*)^{1-\eta}+\alpha(\hat{P}_{Ht}^*)^{1-\eta}=1$$

$$(C_t^*)^{-\sigma}/C_t^{-\sigma}=Q_t$$

- 厂商部门行为方程

$$Y_t=A_tN_t,\quad Y_t^*=A_t^*N_t^*$$

$$\frac{\dot{P}_{Ht}}{P_{Ht}}=\frac{1}{\mu}\frac{S_{1t}}{S_{2t}},\quad \theta\Pi_{Ht}^{\epsilon-1}+(1-\theta)\big(\dot{P}_{Ht}/P_{Ht}\big)^{1-\epsilon}=1$$

$$S_{1t}=(1-\alpha)\hat{P}_{Ht}^{-\eta}C_t^{1-\sigma}\frac{\hat{W}_t}{A_t}+\beta\theta\mathbb{E}_t\Pi_{Ht+1}^{\epsilon}S_{1t+1}$$

$$S_{2t}=(1-\alpha)\hat{P}_{Ht}^{1-\eta}C_t^{1-\sigma}+\beta\theta\mathbb{E}_t\Pi_{Ht+1}^{\epsilon-1}S_{2t+1}$$

$$\frac{\dot{P}_{Ft}^*}{P_{Ft}^*}=\frac{1}{\mu}\frac{X_{1t}^*}{X_{2t}^*},\quad \theta\Pi_{Ft}^{*\epsilon-1}+(1-\theta)\big(\dot{P}_{Ft}^*/P_{Ft}^*\big)^{1-\epsilon}=1$$

$$X_{1t}^*=(1-\alpha)(\hat{P}_{Ft}^*)^{-\eta}(C_t^*)^{1-\sigma}\frac{\hat{W}_t^*}{A_t^*}+\beta\theta\mathbb{E}_t\big(\Pi_{Ft+1}^*\big)^{\epsilon}X_{1t+1}^*$$

$$X_{2t}^*=(1-\alpha)(\hat{P}_{Ft}^*)^{1-\eta}(C_t^*)^{1-\sigma}+\beta\theta\mathbb{E}_t\big(\Pi_{Ft+1}^*\big)^{\epsilon-1}X_{2t+1}^*$$

  - PCP

$$\hat{P}_{Ft}=\hat{P}_{Ft}^*Q_t,\quad \hat{P}_{Ht}^*=\hat{P}_{Ht}/Q_t$$

  - LCP

$$\frac{\dot{P}_{Ht}^*}{P_{Ht}^*}=\frac{1}{\mu}\frac{S_{1t}^*}{S_{2t}^*},\quad \theta\Pi_{Ht}^{*\epsilon-1}+(1-\theta)\big(\dot{P}_{Ht}^*/P_{Ht}^*\big)^{1-\epsilon}=1$$

$$S_{1t}^*=\alpha(\hat{P}_{Ht}^*)^{-\eta}C_t^*C_t^{-\sigma}\hat{W}_t/A_t+\beta\theta\mathbb{E}_t(\Pi_{Ht+1}^*)^{\epsilon}S_{1t+1}^*$$

$$S_{2t}^*=\alpha(\hat{P}_{Ht}^*)^{1-\eta}C_t^*C_t^{-\sigma}Q_t+\beta\theta\mathbb{E}_t(\Pi_{Ht+1}^*)^{\epsilon-1}S_{2t+1}^*$$

$$\frac{\dot{P}_{Ft}}{P_{Ft}}=\frac{1}{\mu}\frac{X_{1t}}{X_{2t}},\quad \theta\Pi_{Ft}^{\epsilon-1}+(1-\theta)\big(\dot{P}_{Ft}/P_{Ft}\big)^{1-\epsilon}=1$$

$$X_{1t}=\alpha\hat{P}_{Ft}^{-\eta}C_t(C_t^*)^{-\sigma}\hat{W}_t^*/A_t^*+\beta\theta\mathbb{E}_t\Pi_{Ft+1}^{\epsilon}X_{1t+1}$$

$$X_{2t}=\alpha\hat{P}_{Ft}^{1-\eta}C_t(C_t^*)^{-\sigma}/Q_t+\beta\theta\mathbb{E}_t\Pi_{Ft+1}^{\epsilon-1}X_{2t+1}$$

- 均衡条件

$$Y_t = (1-\alpha)\hat{P}_{Ht}^{-\eta} C_t V_{Ht} + \alpha(\hat{P}_{Ht}^*)^{-\eta} C_t^* V_{Ht}^*$$

$$Y_t^* = (1-\alpha)(\hat{P}_{Ft}^*)^{-\eta} C_t^* V_{Ft}^* + \alpha \hat{P}_{Ft}^{-\eta} C_t V_{Ft}$$

$$V_{Ht} = \theta \Pi_{Ht}^{\epsilon} V_{Ht-1} + (1-\theta)\left(\dot{P}_{Ht}/P_{Ht}\right)^{-\epsilon}$$

$$V_{Ft}^* = \theta(\Pi_{Ft}^*)^{\epsilon} V_{Ft-1}^* + (1-\theta)\left(\dot{P}_{Ft}^*/P_{Ft}^*\right)^{-\epsilon}$$

  - PCP

$$V_{Ht}^* = V_{Ht}, \quad V_{Ft} = V_{Ft}^*$$

  - LCP

$$V_{Ht}^* = \theta(\Pi_{Ht}^*)^{\epsilon} V_{Ht-1}^* + (1-\theta)\left(\dot{P}_{Ht}^*/P_{Ht}^*\right)^{-\epsilon}$$

$$V_{Ft} = \theta \Pi_{Ft}^{\epsilon} V_{Ft-1} + (1-\theta)\left(\dot{P}_{Ft}/P_{Ft}\right)^{-\epsilon}$$

- 货币政策方程

$$\frac{R_t}{R} = \left(\frac{R_{t-1}}{R}\right)^{\rho_r} \left[\left(\frac{Y_t}{\bar{Y}_t}\right)^{\phi_y} \Pi_{Ht}^{\phi_\pi}\right]^{1-\rho_r}$$

$$\frac{R_t^*}{R^*} = \left(\frac{R_{t-1}^*}{R^*}\right)^{\rho_r} \left[\left(\frac{Y_t^*}{\bar{Y}_t^*}\right)^{\phi_y} (\Pi_{Ft}^*)^{\phi_\pi}\right]^{1-\rho_r}$$

- 定义方程

$$\frac{\hat{P}_{Ht}}{\hat{P}_{Ht-1}} = \frac{\Pi_{Ht}}{\Pi_t}, \quad \frac{\hat{P}_{Ft}^*}{\hat{P}_{Ft-1}^*} = \frac{\Pi_{Ft}^*}{\Pi_t^*}, \quad \frac{\hat{P}_{Ft}}{\hat{P}_{Ft-1}} = \frac{\Pi_{Ft}}{\Pi_t}, \quad \frac{\hat{P}_{Ht}^*}{\hat{P}_{Ht-1}^*} = \frac{\Pi_{Ht}^*}{\Pi_t^*}$$

  - PCP

$$S_t = \frac{\hat{P}_{Ft}^* Q_t}{\hat{P}_{Ht}}, \quad CM_t = 1$$

  - LCP

$$S_t = \frac{\hat{P}_{Ft}}{\hat{P}_{Ht}}, \quad CM_t = \frac{\hat{P}_{Ht}^* Q_t}{\hat{P}_{Ht}}$$

$$\frac{Q_t}{Q_{t-1}} = \Delta\xi_t \frac{\Pi_t^*}{\Pi_t}$$

- 弹性价格下动态方程

$$\bar{C}_t^{\sigma} \bar{Y}_t^{\varphi} = \mu \bar{\hat{P}}_{Ht} A_t^{1+\varphi}, \quad (\bar{C}_t^*)^{\sigma} (\bar{Y}_t^*)^{\varphi} = \mu \bar{\hat{P}}_{Ft}^* (A_t^*)^{1+\varphi}, \quad (\bar{C}_t^*)^{-\sigma} / \bar{C}_t^{-\sigma} = \bar{Q}_t$$

$$\bar{Y}_t = (1-\alpha)\bar{\hat{P}}_{Ht}^{-\eta} \bar{C}_t + \alpha\left(\frac{\bar{\hat{P}}_{Ht}}{\bar{Q}_t}\right)^{-\eta} \bar{C}_t^*$$

$$\bar{Y}_t^* = (1-\alpha)(\bar{\hat{P}}_{Ft}^*)^{-\eta} \bar{C}_t^* + \alpha(\bar{\hat{P}}_{Ft}^* \bar{Q}_t)^{-\eta} \bar{C}_t$$

$$(1-\alpha)\bar{\hat{P}}_{Ht}^{1-\eta} + \alpha(\bar{\hat{P}}_{Ft}^* \bar{Q}_t)^{1-\eta} = 1, \quad (1-\alpha)(\bar{\hat{P}}_{Ft}^*)^{1-\eta} + \alpha\left(\frac{\bar{\hat{P}}_{Ht}}{\bar{Q}_t}\right)^{1-\eta} = 1$$

B 求模型稳态

设稳态时 $\Pi = \Pi^* = 1$，以及 $A = A^* = 1$。稳态时，$\hat{B}_H = \hat{B}^*_{Ht} = 0$；$\Pi_H = \Pi = \Pi^* = \Pi^*_F = 1$；$\hat{P}_H = \hat{P}_F = 1 = \hat{P}^*_H = \hat{P}^*_F$；$Q = 1$；$\hat{R} = \hat{R}^* = 1/\beta$；$V_H = V_F = V^*_H = V^*_F = 1$；$\frac{\dot{P}_H}{P_H} = \frac{\dot{P}^*_F}{P^*_F} = \frac{\dot{P}^*_H}{P^*_H} = \frac{\dot{P}_F}{P_F} = 1$。模型剩余方程构成的稳态系统如下：

- 家庭部门行为方程

$$C^\sigma N^\varphi = \hat{W}, \quad (C^*)^\sigma (N^*)^\varphi = \hat{W}^*$$

- 厂商部门行为方程

$$Y = AN, \quad Y^* = A^* N^*$$

$$S_1 = \mu S_2, \quad S_1 = \frac{1-\alpha}{1-\beta\theta} C^{1-\sigma} \hat{W}, \quad S_2 = \frac{1-\alpha}{1-\beta\theta} C^{1-\sigma}$$

$$X^*_1 = \mu X^*_2, \quad X^*_1 = \frac{1-\alpha}{1-\beta\theta} (C^*)^{1-\sigma} \hat{W}^*, \quad X^*_2 = \frac{1-\alpha}{1-\beta\theta} (C^*)^{1-\sigma}$$

$$S^*_1 = \mu S^*_2, \quad S^*_1 = \frac{\alpha}{1-\beta\theta} C^* C^{-\sigma} \hat{W}, \quad S^*_2 = \frac{\alpha}{1-\beta\theta} C^* C^{-\sigma}$$

$$X_1 = \mu X_2, \quad X_1 = \frac{\alpha}{1-\beta\theta} C (C^*)^{-\sigma} \hat{W}^*, \quad X_2 = \frac{\alpha}{1-\beta\theta} C (C^*)^{-\sigma}$$

- 均衡条件

$$Y = (1-\alpha) C + \alpha C^*, \quad Y^* = (1-\alpha) C^* + \alpha C$$

$$C^* = C$$

求解稳态系统模型稳态均衡为：

$$\hat{W} = \hat{W}^* = \mu, \quad Y = C = N = Y^* = C^* = N^* = \mu^{\frac{1}{\sigma+\varphi}}$$

C 对数线性化

$$c_t = \mathbb{E}_t c_{t+1} - \frac{1}{\sigma}(r_t - \mathbb{E}_t \pi_{t+1})$$

$$c^*_t = \mathbb{E}_t c^*_{t+1} - \frac{1}{\sigma}(r^*_t - \mathbb{E}_t \pi^*_{t+1})$$

$$\sigma c_t + \varphi n_t = \hat{w}_t$$

$$\sigma c^*_t + \varphi n^*_t = \hat{w}^*_t$$

$$(1-\alpha)\hat{p}_{Ht} + \alpha \hat{p}_{Ft} = 0$$

$$(1-\alpha)\hat{p}^*_{Ft} + \alpha \hat{p}^*_{Ht} = 0$$

$$-\sigma(c^*_t - c_t) = q_t$$

$$y_t = a_t + n_t$$

$$y^*_t = a^*_t + n^*_t$$

$$\pi_{Ht} = \beta \mathbb{E}_t \pi_{Ht+1} + \kappa(\hat{w}_t - \hat{p}_{Ht} - a_t)$$

$$\pi^*_{Ft} = \beta \mathbb{E}_t \pi^*_{Ft+1} + \kappa(\hat{w}^*_t - \hat{p}^*_{Ft} - a^*_t)$$

$$\text{PCP:} \hat{p}_{Ft} = \hat{p}^*_{Ft} + q_t, \quad \text{LCP:} \pi_{Ft} = \kappa(\hat{w}^*_t - a^*_t + q_t - \hat{p}_{Ft}) + \beta \mathbb{E}_t \pi_{Ft+1}$$

$$\text{PCP:} \hat{p}^*_{Ht} = \hat{p}_{Ht} - q_t, \quad \text{LCP:} \pi^*_{Ht} = \kappa(\hat{w}_t - a_t - q_t - \hat{p}^*_{Ht}) + \beta \mathbb{E}_t \pi^*_{Ht+1}$$

$$y_t = (1-\alpha)(c_t - \eta\hat{p}_{Ht}) + \alpha(c_t^* - \eta\hat{p}_{Ht}^*)$$
$$y_t^* = (1-\alpha)(c_t^* - \eta\hat{p}_{Ft}^*) + \alpha(c_t - \eta\hat{p}_{Ft})$$
$$r_t = \rho_r r_{t-1} + (1-\rho_r)(\phi_y\tilde{y}_t + \phi_\pi\pi_{Ht})$$
$$r_t^* = \rho_r r_{t-1}^* + (1-\rho_r)(\phi_y\tilde{y}_t^* + \phi_\pi\pi_{Ft}^*)$$
$$\sigma\bar{c}_t + \varphi\bar{y}_t = \bar{\hat{p}}_{Ht} + (1+\varphi)a_t$$
$$\sigma\bar{c}_t^* + \varphi\bar{y}_t^* = \bar{\hat{p}}_{Ft}^* + (1+\varphi)a_t^*$$
$$\bar{y}_t = (1-\alpha)(\bar{c}_t - \eta\bar{\hat{p}}_{Ht}) + \alpha(\bar{c}_t^* - \eta(\bar{\hat{p}}_{Ht} - \bar{q}_t))$$
$$\bar{y}_t^* = (1-\alpha)(\bar{c}_t^* - \eta\bar{\hat{p}}_{Ft}^*) + \alpha(\bar{c}_t - \eta(\bar{\hat{p}}_{Ft}^* + \bar{q}_t))$$
$$(1-\alpha)\bar{\hat{p}}_{Ht} + \alpha(\bar{\hat{p}}_{Ft}^* + \bar{q}_t) = 0$$
$$(1-\alpha)\bar{\hat{p}}_{Ft}^* + \alpha\big(\bar{\hat{p}}_{Ht} - \bar{q}_t\big) = 0$$
$$\sigma(\bar{c}_t - \bar{c}_t^*) = \bar{q}_t$$